

ACG FINAL EVOLUTION REPORT

Universal Agent Commerce Control Plane (Baseline -> V2 -> V3 -> V4)

1. Executive Summary & Core Mission

The **Agent Commerce Gateway (ACG / MACCP)** evolution loop has successfully progressed through all defined milestones:

- **Baseline:** Secure Zero-Trust Transaction Gateway with Cryptographic Mandates and Razorpay Rails.
- **V2 (Control Plane):** Introduction of Agent Principals, Capability Scopes, Policy Decision Point (PDP), Human Confirmation, Hierarchical Budgets, Velocity Controls, Kill Switches, Policy Simulation, and Decision Replay.
- **V3 (Security Infrastructure):** Pluggable Advisory Risk Engine, Granular Decision Traces, Incident Console & Response Workflows, and Property-Based Invariant Verification.
- **V4 (Universal Control Plane):** Universal Authorization API, Capability Negotiation, Multi-Agent Delegation, Policy DSL Compiler, Native MCP Surface, and Payment Rail Abstraction.

Core Thesis:

"The model can propose anything. It cannot authorize anything."

"AI proposes. ACG authorizes. Razorpay executes."

2. Comprehensive Evolution Scorecard

Capability / Subsystem	Baseline	V2 (Control Plane)	V3 (Security Infra)	V4 (Universal Control Plane)
Agent Identity & Principals	DESIGNED	LIVE	LIVE	LIVE
Capability-Based AuthZ	DESIGNED	LIVE	LIVE	LIVE
Policy Decision Point (PDP)	TESTED	LIVE	LIVE	LIVE
Human In-The-Loop Confirmation	DESIGNED	LIVE	LIVE	LIVE
Hierarchical Budget Control	TESTED	LIVE	LIVE	LIVE
Velocity Control Engine	DESIGNED	LIVE	LIVE	LIVE
Global / Merchant Kill Switch	DESIGNED	LIVE	LIVE	LIVE
Policy Simulation Engine	DESIGNED	LIVE	LIVE	LIVE
Deterministic Decision Replay	DESIGNED	LIVE	LIVE	LIVE
Finite Financial State Machine	TESTED	LIVE	LIVE	LIVE
Pluggable Risk Provider	DESIGNED	DESIGNED	LIVE	LIVE
Granular Decision Tracing	DESIGNED	DESIGNED	LIVE	LIVE
Agent Incident Console & Ops	DESIGNED	DESIGNED	LIVE	LIVE
Property-Based Invariant Tests	DESIGNED	DESIGNED	LIVE	LIVE
Chaos & Fail-Closed Safety	TESTED	TESTED	LIVE	LIVE
Universal Authorization API	DESIGNED	DESIGNED	DESIGNED	LIVE
Canonical Financial IR	TESTED	TESTED	TESTED	LIVE
Capability Negotiation	DESIGNED	DESIGNED	DESIGNED	LIVE
Multi-Agent Delegation	DESIGNED	DESIGNED	DESIGNED	LIVE
Agent Trust Lifecycle	DESIGNED	TESTED	TESTED	LIVE
Policy Compiler & Validator	DESIGNED	DESIGNED	DESIGNED	LIVE
Cross-Protocol Conformance	TESTED	TESTED	TESTED	LIVE
Payment Rail Abstraction	TESTED	TESTED	TESTED	LIVE
Distributed Architecture Target	PRODUCTION TARGET	PRODUCTION TARGET	PRODUCTION TARGET	PRODUCTION TARGET

3. Quantitative Verification Results

- Automated Test Suite: 102 / 102 Passed (12 Test Suites)
- Baseline Gateway & Adapters: 77 Tests
- V2 Agent Financial Control Plane: 10 Tests
- V3 Agent Security Infrastructure: 7 Tests

- V4 Universal Control Plane: 8 Tests
 - **Adversarial / Pentest Scenarios:** 19 / 19 Passed (0 bypasses, 0 regressions)
 - **Unauthorized Financial Paths Observed:** 0
 - **Tamper-Evident SHA-256 Audit Ledger:** 307 Blocks Cryptographically Sound
 - **End-to-End Cold Run Benchmark:** 306.17 ms (Clean cold boot to Razorpay Order creation)
 - **Merchant Integration Benchmark:** ~10-12 minutes
 - **Production Build:** TypeScript & Vite clean build passing in 3.30s
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4. Known Production Boundaries & Next Steps

- **Current Runtime Boundary:** Single-node SQLite reference implementation with atomic transactions and in-memory test harnesses.
- **Production Architecture Target:** Multi-instance distributed deployment backed by PostgreSQL, durable transactional outbox, worker queues, and HSM/KMS-managed signing keys as detailed in `docs/production/SCALING_ARCHITECTURE.md`.